Ruthenium-106 plaque brachytherapy for uveal melanoma

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ABSTRACT

Background To report on local tumour control, eye preservation and visual outcome after ruthenium-106 brachytherapy for uveal melanoma.

Methods Medical records of 143 eyes with uveal melanoma, treated by ruthenium-106 brachytherapy between 1997 and 2012 at one single centre, were included. Primary outcome measures were local tumour control, eye preservation and visual outcome. The influence of patient, tumour and treatment parameters on outcome was analysed by time to event analysis and competing risk regression.

Results The median overall follow-up was 37.9 months. Tumour control: recurrent tumour growth was observed in 17 patients. The estimated local tumour recurrence rate at 12, 24 and 48 months after irradiation was 3%, 8.4% and 14.7%, respectively. The only significant risk factors for tumour recurrence were age (p=0.046) and reduced initial visual acuity (VA, p=0.045). No significant difference could be shown for tumour size or tumour category (T1–T2 vs T3–T4), and for any other tumour or treatment parameters (including combined transpupillary thermo-therapy (TTT)).

Eye preservation: The likelihood of keeping the eye 12, 24 and 48 months after irradiation was 97.7%, 94.7% and 91.8%, respectively. Most significant risk factors for secondary enucleation were initial VA (p<0.001), tumour height (p=0.002) and tumour category (p=0.015).

Vision The chances of keeping VA of 20/200 or better at 1, 2 and 5 years after treatment were 86.4%, 80.8% and 61.7%, respectively. Patients receiving sandwich-TTT showed significantly worse visual outcomes.

Conclusions Ruthenium-106 brachytherapy appears to be a useful treatment regarding tumour control, eye preservation and visual function. Adjunct sandwich therapy resulted in worse visual outcome.

INTRODUCTION

Over the past decades, treatment of uveal melanoma has changed significantly. Since the Collaborative Ocular Melanoma Study (COMS) has demonstrated equal survival rates for enucleation and plaque brachytherapy for medium-sized melanomas, eye-preserving therapies have largely replaced enucleation as primary treatment, and radiotherapeutic treatment modalities have become the mainstay in uveal melanoma therapy.[1 2] Today, the occurrence of metastasis and patient survival is generally thought to be independent of the treatment modality. The spread of tumour cells appears to be an early event in the development of uveal melanoma, and the development of metastatic disease seems to depend primarily on intrinsic tumour characteristics, such as epitheloid cell type, extravascular matrix patterns, cytogenetic changes and class 1 or class 2 gene expression profile.[3] As local treatment cannot prevent metastasis, the main goals of treatment are local tumour control, preservation of the eye and the conservation of some useful visual function.

Besides proton beam radiotherapy, which is performed in only a few specialised ophthalmic oncology centres, brachytherapy is the most commonly used form of radiotherapy for uveal melanomas. Since iodine-125 plaques have been used in the COMS, this isotope may be considered the gold-standard for plaque brachytherapy. However, the use of other isotopes, such as palladium-103 and ruthenium-106, has been reported in smaller studies.[4–14] Ruthenium-106 is a β-emitter, with a steep dose fall-off, and lower penetration depth compared with iodine-125.[15] Tumours with a height of >6 mm are generally considered as unsuited for ruthenium-106 plaque therapy.

Nevertheless, some authors published promising results for larger tumours reporting high local tumour control and preservation of vision even in tumours >8 mm.[8] However, results are generally worse than those reported for smaller tumours.[16 17]

We evaluated visual function, local tumour control and eye preservation after ruthenium-106 brachytherapy of uveal melanomas and assessed the impact of tumour and patient parameters on treatment outcome.

PATIENTS AND METHODS

Study design

Initially 172 records were identified and reviewed. Twelve patients were excluded because the melanoma had previously been treated by other treatment modalities (11 cases transpupillary thermo-therapy (TTT); 1 case with gamma knife radiosurgery). Isolated iris melanomas (11 patients) and six cases with incomplete medical records were also excluded. In total, 143 eyes of 143 patients were included into our analysis. Out of these, 24 patients (16.78%) had planned combined adjunct TTT ('sandwich therapy') 3 months before or after plaque brachytherapy. Plaque therapy was offered to patients as a globe-preserving treatment option; if tumour characteristics allowed local brachytherapy, patients agreed to repeated follow-up examinations and gave informed consent. Exclusion criteria for plaque brachytherapy included patient´s refusal of any invasive treatment and/or follow-up visits. Large tumour size (height >7 mm, longest basal diameter >25 mm) and a posterior tumour margin

Table 1 Overview about patient, tumour and treatment characteristics

Characteristic	Value (%)
Sex	
Male	62 (43.4)
Female	81 (56.6)
Laterality	
Right	71 (49.6)
Left	72 (50.3)
Age (years)	62.50 (median) (IQR 53.4–74.0; range 28.6–88.7)
Follow-up (months)	37.88 (median; IQR 20.37–62.0)
Systemic diseases	
Systemic diabetes	16 (11.2)
Systemic hypertension	60 (42.0)
Coronary artery disease	14 (9.8)
Other systemic cancer	13 (9.1)
Ocular abnormalities of the affected eye	
Retinal detachment	68 (47.5)
Cataract	59 (41.2)
Pseudophakia or aphakia	13 (9.1)
Glaucoma	4 (2.8)
Macular degeneration	7 (4.9)
Diabetic retinopathy	1 (0.7)
Choroidal nevus	3 (2.1)
Tumour parameters	
Tumour height (mm)	4.5 (median) (IQR 3.8–6.6; range 2.2–13.4)
<6	98 (68.5)
≥6	45 (31.5)
Longest basal diameter (mm)	11 (median) (IQR 9.5–13.4; range 4.3–20.5)
Tumour distance to disc (mm)	6 (median) (IQR 3.7–9.9; range 0–21.8)
Tumour distance to fovea (mm)	7.5 (median) (IQR 3.7–10.5; range 0–21.3)
TNM	
T1	36 (25.2)
T2	70 (48.9)
T3	32 (22.4)
T4	5 (3.5)
Shape	
Dome (114) or plateau (4)	118 (82.5)
Mushroom	25 (17.5)
Tumour pigmentation	
Melanotic	132 (92.3)
Amelanotic	11 (7.7)
Coronal location	
Nasal	56 (39.2)
Midline	29 (20.3)
Temporal	58 (40.6)
Sagittal location	
Superior	60 (42.0)
Horizontal	20 (14.0)
Inferior	63 (44.0)
Anterior margin	
Ciliary body	28 (19.6)
Anterior to equator	79 (55.2)
Posterior to equator	36 (25.2)
Posterior margin	
≤1.5 mm from optic disc or fovea	16 (11.2)
1.6–3.0 mm from optic disc or fovea	31 (21.7)
Posterior to equator; ≥3 mm from optic disc or fovea	84 (58.7)
Anterior to equator	12 (8.4)
Initial visual acuity	
20/40	105 (73.4)
20/200	138 (96.5)
CF	141 (98.6)
Treatment parameters	
AD (Gy)	89.8 (median) (IQR 59.3–120.8; range 2.5–260.7)
AD <80 Gy	60 (42)
AD 80–110 Gy	37 (26)
AD >110 Gy	46 (32)
SSD (Gy)	689.8 (median) (IQR 488.7–922.8; range 189.2–1414.7)
SSD <200 Gy	1 (0.7)
SSD 200–500 Gy	37 (26)
SSD 500–800 Gy	47 (33)
SSD >800 Gy	58 (41)

Depending on tumour basal diameter the following plaques were used: CCA (PD: 15.3 mm), CCB (PD: 20.2 mm), CCC (PD: 24.8 mm), CIB (PD: 20.2 mm), CCX (PD: 11.6 mm), COB (PD: 19.8 mm), manufactured by BEBIG, Berlin, Germany. Visual acuity was measured in Snellen lines (20/20, 20/25, 20/30, 20/40, 20/50, 20/60, 20/70, 20/80, 20/100, 20/125, 20/160, 20/200, 20/400, counting fingers, hand motions, light perception, no light perception).
AD, radiation dose to apex; CF, counting fingers; PD, plaque diameter; SSD, scleral surface radiation dose; T, tumour category; TNM, tumour nodes metastasis category.

within 1.5 mm from the optic disc or from the centre of the fovea were relative contraindications, and the patients were informed explicitly about the possible adverse outcome of the treatment in the presence of those risk factors.

Pre-irradiation assessment

At initial presentation, a full ophthalmic exam was performed (visual acuity (VA), slit-lamp and indirect fundus examination) and fundus photographs were taken. In addition, tumour size was measured by standardised echography (A/B-scan) in all cases. Optical coherent tomography and ocular angiography (fluorescein and indocyanine green) were performed if necessary. We also assessed patient demographics and past medical and family history. All data had been recorded prospectively at time of examination in an electronic medical record and database (FileMaker, V.3–6 and EyMed). The variables evaluated for the present study are described in detail in table 1.

Follow-up

The first follow-up examination was done approximately one month after plaque removal and thereafter scheduled every three months for the first two years and every six months up to five years after irradiation. Thereafter patients were followed annually. The ophthalmic exam was repeated at each visit. To rule out metastatic disease, systemic examinations were performed, including blood work with liver parameters and abdominal ultrasound examination every half a year, as well as chest X-ray examinations annually.

Treatment protocol

The intraocular tumours in our study were treated with radioactive ruthenium-106 applicators of different size, depending on tumour basal diameter. Treatment was aimed to deliver a minimal radiation dose of 80–100 Gy to the tumour apex, not exceeding a maximal scleral dose of 1000 Gy. In selected cases of larger tumours, higher scleral surface doses were accepted.[5] Surgery was usually performed in general anaesthesia for plaque insertion and positioning. During surgery, transpupillary transillumination, scleral indentation, the application of a transparent plastic template and indirect ophthalmoscopy were used to verify tumour margins and to ensure proper positioning of the radioactive plaque. The plaque was positioned to cover a minimal safety margin of 1 mm around the tumour margins. External ocular rectus and oblique muscles were disinserted, if necessary, and reattached after treatment (rectus muscles only). In general, plaques were removed in retrobulbar anaesthesia.

Outcome measures

Primary outcome measures were local tumour control, eye preservation and visual outcome.

Tumour recurrence was defined as (1) an increase in tumour height over two consecutive follow-up examinations at least six months after irradiation or (2) an increase of the tumour basal diameter compared with preoperatively taken fundus photographs any time after irradiation.

For the analysis of eye retention, the time to secondary enucleation was calculated and the reason for the loss of the eye was evaluated. The end of follow-up was defined as the day of the last presentation of the patient at our department or as the day of enucleation of the affected eye regardless to causality.

Statistical analysis

Statistical analysis was performed by R V.2.14.2 and Stata V.13.0. The values of categorical variables are presented as relative and absolute frequencies, continuous variables as median, range and IQR. Follow-up was calculated by the inverse Kaplan–Meier method.

Cumulative incidence function (CIF) was used to describe the occurrence of tumour recurrence over time after treatment in the presence of enucleation as a competing risk. Gray's test was performed for the comparison of CIFs between groups, and competing risk regression (Fine–Gray method) was used for the analysis of risk factors for tumour recurrence.

The risk for secondary enucleation and visual loss was estimated using the Kaplan–Meier method. The time to visual loss was defined as the midpoint between the first time of visual loss and the previous exam. Cox's univariate proportional hazard model was used to evaluate possible risk factors. For each variable, HRs were calculated. A stepwise multivariate Cox regression analysis including those variables that showed a p value of ≤0.05 in the univariate procedure was used to identify baseline variables as independent risk factors for visual deterioration. All variables were checked for collinearity; p values of ≤0.05 were defined as statistically significant.

RESULTS

Ocular status before treatment

VA: Best-corrected visual function on the affected eye before treatment was 20/40 or better in 105 patients, equal to or better than 20/200 in 139 patients and equal to or better than counting fingers (CF) in 141 patients. Pre-irradiation patient and tumour characteristics are summarised in table 1.

The median overall follow-up, calculated by the inverse Kaplan–Meier method, was 37.9 months (median; IQR 20.4–62.0).

Treatment characteristics

The median dose delivered to the tumour apex was 89.8 Gy (IQR 59.3–120.8; range 2.5–260.7). The median dose to the external sclera was 689.8 Gy (IQR 488.7–922.8; range 169.2–1414.7). Adjunct sandwich therapy was planned and performed in case of uncertain irradiation sufficiency, mainly in case of extended tumour height (24 cases; first application was performed in April 1997).

Local tumour control

In total, 17 (11.9%) of 143 eyes showed recurrent tumour growth. The estimated local tumour recurrence rate at 12, 24 and 48 months after irradiation was 3% (95% CI 1.0% to 7.1%), 8.4% (95% CI 4.3% to 14.3%) and 14.7% (95% CI 8.5% to 22.6%), respectively, calculated by a competing risk regression (Fine–Gray model, figure 1). No significant difference in local control could be shown between large and small intraocular melanoma (T1–T2 vs T3–T4). No association was found between the difference of longest tumour basal diameter and applicator diameter, and local tumour recurrence. Local tumour recurrence after 12, 24 and 48 months in small melanoma was 4.1% (95% CI 1.3% to 9.5%), 8.8% (95% CI 4.1% to 15.7%) and 12.0% (95% CI 6.0% to 20.2%), and 0%, 7.5% (95% CI 1.3% to 21.4%) and 23.5% (95% CI 8.3% to 43.1%) for large melanoma. Univariate risk factors for local tumour recurrence are summarised in table 2.

Eye preservation

In total, 10 (7%) eyes had to be removed after brachytherapy, 8 eyes because of tumour recurrence, 1 for neovascular glaucoma and 1 for persisting choroidal detachment. The estimated probability (Kaplan–Meier method) of keeping the eye 12, 24 and 48 months after irradiation was 97.7% (95% CI 93.0% to 99.3%), 94.7% (95% CI 88.6% to 97.6%) and 91.8% (95% CI 83.9% to 95.9%), respectively (figure 1). The most important univariate risk factors for secondary enucleation were a reduced initial VA, large tumour size, male gender and low dose to tumour apex. Univariate risk factor analyses for enucleation are summarised in table 2.

Functional outcome

At the end of follow-up, VA in the affected eye was equal to or better than 20/40, 20/200 or CF in 56, 100 and 117 cases.

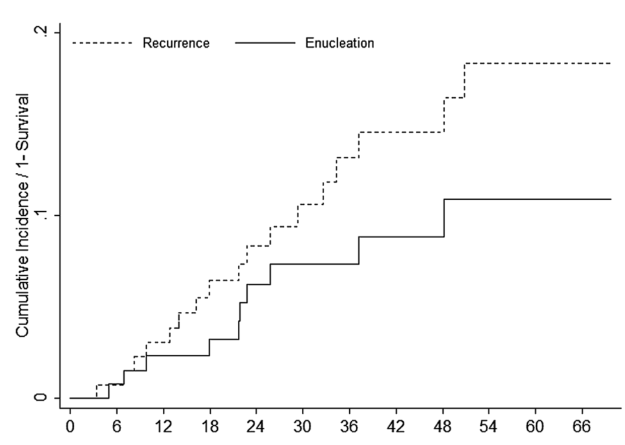

Figure 1 Probability of tumour recurrence and enucleation: cumulative incidence over time after treatment.

Table 2 Results of the univariate risk analysis for local tumour recurrence and eye preservation

Univariate analysis	SHR	95% CI for HR Lower	95% CI for HR Upper	p Value
Local tumour recurrence				
Initial VA	0.9	0.8	1	0.045*
T-height	1.1	0.9	1.4	0.190
LBD	1.1	1.0	1.3	0.086
Tumour size (T3–T4)	1.7	0.6	4.6	0.285
Distance to disc ≥3 mm	1.3	0.3	5.7	0.763
Distance to fovea ≥3 mm	1.6	0.3	7.7	0.523
Dose to apex <90 Gy	2.9	1.0	9.1	0.061
Age	1.0	1.0	1.1	0.046*
Gender (male)	2.0	0.8	5.1	0.158
Eye preservation/secondary enucleation				
Initial VA	0.8	0.7	0.9	<0.001*
T-height	1.4	1.1	1.7	0.002*
LBD	1.1	0.9	1.4	0.170
Tumour size (T3–T4)	4.7	1.3	17.0	0.015*
Distance to disc ≥3 mm	1.6	0.2	12.9	0.644
Distance to fovea ≥3 mm	0.4	0.1	1.7	0.249
Dose to apex <90 Gy	7.9	1.0	62.3	0.051
Age	1.0	1.0	1.1	0.265
Gender (male)	5.6	1.2	26.7	0.029*

Multivariate risk analysis was not performed because the number of recurrent tumour growths and enucleated cases was too small (n=17, n=10). Values marked by'*' were found to be statistically significant.
LBD, longest basal tumour diameter; SHR, sub-HR; T-height, tumour height; VA, visual acuity.

The chances of keeping good VA of 20/40 at 1, 2 and 5 years after treatment were 72.74% (95% CI 62.99% to 80.31%), 64.85% (95% CI 54.5% to 73.4%) and 44.39% (95% CI 32.82% to 55.32%), respectively. The chances of keeping VA of 20/200 at 1, 2 and 5 years after treatment were 86.42% (95% CI 79.29% to 91.23%), 80.83% (95% CI 72.66% to 86.77%) and 61.77% (95% CI 50.15% to 71.44%), respectively (figure 2). The chances of keeping VA of CF 1, 2 and 5 years after treatment were 93.91% (95% CI 88.18% to 96.91%), 90.27% (95% CI 83.43% to 94.38%) and 74.36% (95% CI 62.98% to 82.72%). Univariate and multivariate risk factors predicting visual loss are summarised in table 3.

In univariate analysis, five risk factors for vision loss were found for all three levels of VA (20/40, 20/200, CF): tumour height, tumour shape, tumour size (T3–T4), initial VA and adjunct TTT. In a direct comparison of visual outcome between cases treated with brachytherapy alone and cases, treated with combined adjunct TTT (n=24), visual outcome was worse in all analysed levels 20/40, p=0.0156; 20/200, p=0.0023; CF, p=0.0018 (log-rank test). However, the use of adjunct TTT was associated with advanced tumour height and a lower dose to tumour apex, and did not remain significant in multivariate analysis.

Tumours >7 mm

As many ocular oncologists are reluctant to treat tumours >7 mm with ruthenium-106 brachytherapy, we analysed this subgroup (n=30) separately. Recurrence rate (n=6) was 11.8% (95% CI 2.9% to 27.3%) and 21.4% (95% CI 7.6% to 39.6%) at 24 and 48 months after treatment. Enucleation rate (n=7) was 3.3% (95% CI 0.5% to 21.4%), 15.4% (95% CI 6.1% to 36.2%) and 26.1% (95% CI 12.4% to 50%) at 12, 24 and 48 months after treatment. Tumour height ≥7 mm was a significant risk factor for secondary enucleation (p=0.004, HR 7.5, 95% CI 1.9 to 29.2), but not for tumour recurrence (p=0.231, sub-HR: 1.8, 95% CI 0.68 to 4.7). A tumour height ≥7 mm was also a significant risk factor for loss of VA of 20/40 (p<0.001; HR 3.6, 95% CI 2.0 to 6.6), 20/200 (p>0.001; HR 3.3, 95% CI 1.7 to 6.3) and CF (p=0.001; HR 4.2, 95% CI 1.9 to 9.5).

DISCUSSION

Since the first description of ruthenium-106 plaque brachytherapy for uveal melanomas by Lommatzsch,[18] a number of reports have been published on this technique.[5–14] Most of those report on the treatment of small-sized and medium-sized melanomas, as recently published by Salkola *et al*.[19]

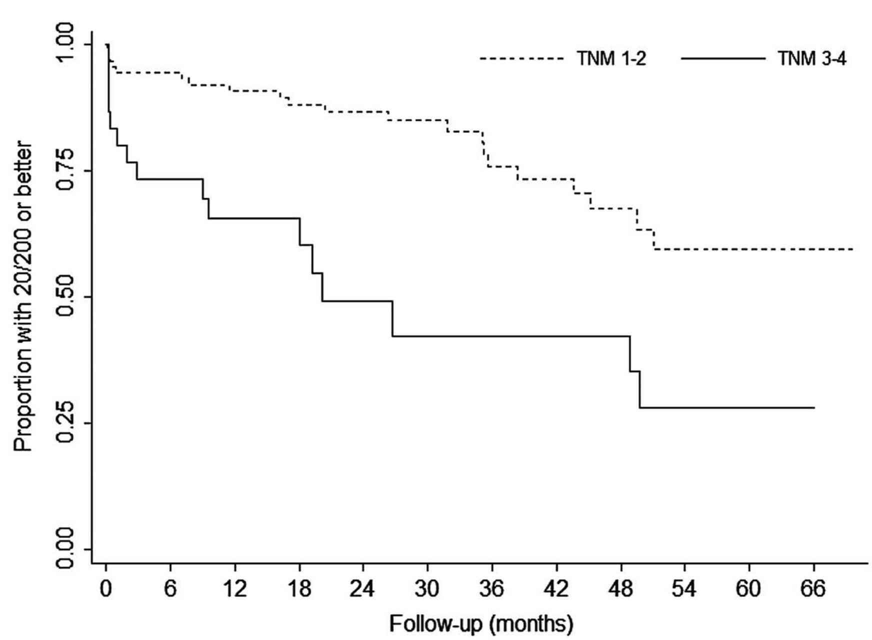

Figure 2 Probability for keeping an initial visual acuity for 20/200 or better after treatment: risk of visual loss according to tumour category (T1/T2 vs T3/T4; Kaplan–Meier estimates). TNM, tumour nodes metastasis.

Table 3 Results of the univariate and multivariate risk analysis for visual loss

	HR	95% CI for HR Lower	Upper	p Value
Univariate analysis 20/40				
T-height	1.4	1.2	1.5	<0.001*
LBD	1.2	1.1	1.3	<0.001*
Shape	0.3	0.2	0.6	0.001*
T3–T4	3.0	1.7	5.5	<0.001*
TTT sandwich	2.2	1.1	4.2	0.019*
Distance to disc >=3 mm	0.3	0.1	0.7	0.005*
Distance to fovea >=3 mm	0.6	0.3	1.2	0.154
Initial VA	0.7	0.5	0.9	0.019*
Initial RD	2.9	1.6	5.0	<0.001*
Multivariate analysis 20/40				
T-height	1.2	1.1	1.4	0.004*
Initial RD	1.9	1.1	3.6	0.031*
Distance to disc >=3 mm	0.4	0.2	0.8	0.014*
Univariate analysis 20/200				
T-height	1.2	1.1	1.4	<0.001*
LBD	1.1	1.0	1.2	0.018*
Shape	0.4	0.2	0.9	0.021*
T3–T4	2.8	1.4	5.3	0.002*
TTT sandwich	2.7	1.4	5.3	0.003*
Distance to disc >=3 mm	0.3	0.1	0.6	0.002*
Distance to fovea >=3 mm	0.9	0.4	2.0	0.721
Initial VA	0.8	0.7	0.9	<0.001*
Initial RD	2.5	1.3	4.9	0.007*
Multivariate analysis 20/200				
T-height	1.2	1.1	1.4	0.001*
Distance to disc >=3 mm	0.3	0.1	0.7	0.003*
Initial VA	0.9	0.8	1.0	0.020*
Univariate analysis CF				
T-height	1.3	1.1	1.4	0.001*
LBD	1.1	1.0	1.2	0.143
Shape	0.3	0.1	0.7	0.005*
T3–4	3.8	1.7	8.6	0.001*
TTT sandwich	3.4	1.5	7.7	0.003*
Distance to disc >=3 mm	0.6	0.2	1.5	0.255
Distance to fovea >=3 mm	1.0	0.3	3.0	0.954
Initial VA	0.9	0.8	1.0	0.030*
Initial RD	1.7	0.8	3.9	0.186
Multivariate analysis CF				
T-height	1.3	1.1	1.4	0.001*

Values marked by '*' were found to be statistically significant. All variables describing tumour parameters are presented in the table (tumour size, shape, expansion and topography). In addition, variables found statistically significant in one vision level are presented for all levels.
CF, counting fingers; LBD, longest basal tumour diameter; RD, retinal detachment; Shape, tumour shape (dome and plateau, mushroom); T3–T4, tumour category T3–T4; T-height, tumour height; TTT, combination of TTT with brachytherapy (sandwich-therapy); VA, visual acuity.

In our study, we analysed the outcome of ruthenium-106 brachytherapy for a wide spectrum of uveal melanomas, including tumours >6 mm (n=46, 32%), the upper limit recommended by the American brachytherapy society for this isotope.[20]

Recurrence

The overall local tumour recurrence in our study was 3.0%, 8.4% and 14.7% at 12, 24 and 48 months after irradiation, respectively. This is well in the range of local tumour control rates reported in the literature.[21] Some authors describe excellent results with recurrence occurring in 3–4% of patients.[6 12] Others report significantly higher recurrence rates up to 16%.[11 22 23] Those differences might be explained by differences in tumour size and location as higher recurrence rates tend to be observed in larger tumours and tumours located posteriorly.[6 10 19] However, in our analysis we failed to show a statistically significant influence of distance to optic disc or fovea on recurrence and eye preservation (table 3).

We also could not find any difference in tumour control between large (tumour nodes metastasis (TNM)3–4) and small uveal melanomas (TNM1–2), or tumour height, suggesting that ruthenium-106 plaque brachytherapy could be used for larger tumours in appropriate indications, as reported previously by Kaiserman *et al*,[8] though recurrence rates were higher with lower doses to the tumour apex. In our series, patients receiving a tumour apex dose of <90 Gy were also more likely to encounter tumour recurrence, yet the difference was not quite significant (p=0.06). Other authors did not find any association between apex dose and tumour recurrence or secondary enucleation; however, dose rate might also play a role in this context.[10 24] The use of combined brachytherapy and TTT for the treatment of melanomas >6 mm has also been described to increase tumour control, and TTT has also been used to reduce radiation dose in case of small tumours or in case of uncertain tumour regression after treatment.[12] Using this approach, Verschueren *et al*[12] and Damato *et al*[6] achieved very high tumour control rates (about 96% and 98% 5 years after treatment). In our series, adjunct TTT did not influence tumour recurrence, but the number of patients was low (24 cases with adjunct sandwich therapy; 17 cases of tumour recurrence, 4 of those after sandwich therapy).

Visual function

Besides successful eye retention, another important factor for the patients remains the preservation of useful vision. The reported 5-year probabilities for keeping VA of 20/40 or better range in between 31% and 74.9%,[5 17 25] and between 49% and 92%[5] for keeping VA 20/200.[8 12 17] Keeping CF or better has been achieved in 97% and 83% at 5 and 9 years after treatment in one series, respectively.[17] Results of our series are in those reported ranges, with probabilities of keeping 20/40, 20/200 and CF 5 years after treatment of 47.3%, 60.7% and 74.0%, respectively. In univariate analysis, five factors for vision loss were found significant for all three levels of VA (20/40, 20/200, CF): tumour height, tumour shape, tumour size (T3–T4), initial VA and adjunct TTT. On multivariate analysis, only tumour height was a significant risk factor for visual loss in all three levels.

The impact of adjunct TTT on visual outcome after brachytherapy is currently under discussion. Damato *et al* did not report any differences in case of adjunct TTT for the probability of keeping VA; however, this group used TTT just in case of eccentric plaque positioning, or if radiation effect was unsure, whereas Verschueren *et al*, using TTT also in case of increased tumour height, found patients receiving adjunct TTT to have reduced probabilities in keeping visual function. We found adjunct TTT to result in significantly worse visual outcome in all analysed levels (20/40, p=0.0156; 20/200, p=0.0023; CF, p=0.0018); however, the use of TTT was associated with advanced tumour height.

Limitations of our study were the relatively small number of patients with tumour recurrence and TTT, given the wide variety of tumours included in our study. Still, our results suggest that ruthenium-106 brachytherapy is a useful treatment in patients with uveal melanomas (even >6 mm) achieving a high eye retention rate and preservation of useful visual function in many cases. Furthermore, our results suggest that the application of adjunct TTT should be considered carefully as this might be associated with worse visual outcome.

Contributors All authors substantially contributed to preparation of the manuscript (according to ICMJE requirements). In detail, their authorship is based on contribution to LT and WW: conception and design of the study, data acquisition, analysis and interpretation of data, drafting and revising the manuscript. CM, PW, and MS: data acquisition, drafting and revising the manuscript. AA: conception and design of the study, analysis and interpretation of data, and revising the manuscript. GL: conception and design of the study, interpreting the results and revising the manuscript.

Competing interests None declared.

Ethics approval Institutional review board approval was obtained to retrospectively analyse medical records of patients treated with ruthenium-106 brachytherapy between June 1997 and July 2012 at the department of ophthalmology at Medical University of Graz (IRB00002556).

Provenance and peer review Not commissioned; externally peer reviewed.

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